NUMBERS THAT ARE BOTH CARMICHAEL AND LUCAS–CARMICHAEL: A VERIFIED BOUND OF 10^{24}, THE FOUR-PRIME CASE, AND OBSTRUCTIONS TO CONSTRUCTIONS

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ABSTRACT. A squarefree composite n is *Carmichael* if $(p-1) \mid (n-1)$ for every prime $p \mid n$, and *Lucas–Carmichael* if $(p+1) \mid (n+1)$ for every $p \mid n$. Williams (1977) asked whether a number can be both; no example is known, and the only published bound is Pinch's 10^{16} (2007). Such numbers are also the backbone of the standard construction of a Baillie–PSW counterexample and of the Lenstra–Pomerance route to a counterexample to Agrawal's conjecture, although those literatures do not use the name. We verify, independently of the tabulators' own flags, that no Carmichael number below 10^{24} is Lucas–Carmichael (308,279,939 numbers; the best near miss has four of nine primes satisfying $(p+1) \mid (n+1)$). For the first open case after Williams's theorem that three prime factors are impossible, we prove that a four-prime example $p_1 < p_2 < p_3 < p_4$ must have all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{2p_1p_2p_3}$ and $p_2^2 \le 24p_1^3+1$, so that only finitely many candidates exist for each smallest prime, and we show by a complete enumeration that none exists with $n \le 10^{30}$ and none at all with $p_1 \le 500{,}000$. We also prove that the two classical construction paradigms for Carmichael numbers fail in their simplest two-sided forms: no Chernick-type polynomial family of three factors, of any number of factors with uniform offsets ± 1, or in generic position can produce such numbers, and we quantify the failure of the two-sided Erdős subset-product method. Heuristics suggest the first example, if any, has hundreds of digits and around a hundred prime factors.

1. INTRODUCTION

Throughout, n denotes an odd squarefree composite integer with prime factorisation $n = p_1p_2\cdots p_k$, $p_1 < \cdots < p_k$, $k = \omega(n) \ge 2$.

Definition 1.1. n is a *Carmichael number* if $(p-1) \mid (n-1)$ for every $p \mid n$ (Korselt's criterion), and a *Lucas–Carmichael number* if $(p+1) \mid (n+1)$ for every $p \mid n$. We call n a *CLC number* if it is both.

Carmichael numbers are the absolute Fermat pseudoprimes; Lucas–Carmichael numbers play the same role for Lucas sequences with discriminant a quadratic non-residue of every $p \mid n$. H. C. Williams [20] asked in 1977 whether a number can be both, proved that it cannot have exactly three prime factors and cannot be divisible by 3, and showed that under his Lucas-sequence formulation the number of prime factors must be odd (in the bare divisibility formulation above, all p_i lie in one class modulo 4, and k is odd when that class is 3; see Lemma 2.2). No example is known. Pinch [14] reported that there is none up to 10^{16}; to our knowledge no other bound and no further theorem has appeared since 1977.

The question is not isolated. Pomerance's 1984 heuristic argument that counterexamples to the Baillie–PSW primality test exist [16], and the subsequent attempts to construct one [4, 3, 10], seek $n = \prod p$ with $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for all $p \mid n$ together with side conditions ($p \equiv 3 \pmod 8$, $(5/p) = -1$); such an n is a CLC number. The Lenstra–Pomerance analysis of Agrawal's conjecture from the AKS primality test [1, 13] shows that the natural counterexamples

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are CLC numbers whose primes are all $\equiv 3 \pmod{80}$. Howe's higher-order Carmichael numbers [11] are a third neighbour: n is Carmichael of order 2 iff $n \equiv p \pmod{p^2-1}$ for all $p \mid n$, while n is CLC iff $n \equiv p \pmod{(p^2-1)/2}$ (Lemma 2.1). These three literatures do not cite one another and do not use a common name; one purpose of this note is to put them side by side.

Our results are the following.

- **Verified bound** (Theorem 4.1). No CLC number is smaller than 10^{24}. This is a complete, independent factor-and-test pass over the tabulations of all Carmichael numbers below 10^{24}, with completeness and per-class cross-checks against independently published counts. The largest number of primes $p \mid n$ with $(p+1) \mid (n+1)$ attained below 10^{24} is four (once, with $\omega(n) = 9$); 826 Carmichael numbers attain three.
- **Four prime factors** (Theorems 3.1 and 3.6). If $n = p_1p_2p_3p_4$ is CLC then all $p_i \equiv 1 \pmod{12}$, $p_3 < 2p_1p_2$, $p_4 < \sqrt{2p_1p_2p_3}$, and $p_2^2 \le 24p_1^3 + 1$; in particular $n < 477\,p_1^{15/2}$ and for each p_1 there are finitely many candidates. A complete enumeration shows that no four-prime CLC number has $n \le 10^{30}$, and that none exists with $p_1 \le 500{,}000$.
- **Construction obstructions** (Theorems 5.1–5.3, §5.2). Chernick-type polynomial families cannot produce CLC numbers with three factors, with uniform offsets ± 1, or in generic position, and the degenerate strata are empty in the ranges searched. The two-sided Erdős–Alford–Granville–Pomerance subset-product method is quantified to lose about 1.5 bits per prime relative to the one-sided case, with the expected number of solutions in the best configurations found near 2^{-30}.
- **Heuristics** (§6). The expected number of four-prime CLC numbers is about 10^{-6}, concentrated at $p_1 = 13$; an Erdős–Pomerance-style count suggests the first CLC number of any kind has between roughly 200 and 500 digits and about 80–110 prime factors.

All code, logs and result files are public (§7). Every computational claim was re-verified by a second, independent route described there.

2. Preliminaries

For an odd prime p write $L_p = (p^2-1)/2 = \operatorname{lcm}(p-1, p+1)$.

Lemma 2.1. *Let $p \mid n$. Then $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ hold together if and only if $n/p \equiv 1 \pmod{L_p}$, equivalently $n \equiv p \pmod{L_p}$.*

Proof. The two divisibilities say $n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1}$. Since $p \equiv 1 \pmod{p-1}$ and $p \equiv -1 \pmod{p+1}$, this is $n \equiv p$ modulo $\operatorname{lcm}(p-1, p+1) = L_p$ (the gcd of $p \mp 1$ is 2). Finally $L_p \mid n - p = p\,(n/p - 1)$ with $\gcd(p, L_p) = 1$ gives $L_p \mid n/p - 1$, and conversely. □

Lemma 2.2 (congruence classes)**.** *Let n be CLC with k prime factors.*

(a) *All $p \mid n$ lie in one class c modulo 4, and $n \equiv c \pmod 4$; hence k is odd if $c = 3$.*
(b) *$3 \nmid n$; all $p \mid n$ lie in one class c' modulo 3, and $n \equiv c' \pmod 3$; hence k is odd if $c' = 2$.*
(c) *If k is even, every $p \mid n$ is $\equiv 1 \pmod{12}$.*

Proof. (a) $8 \mid p^2 - 1$ for odd p, so $4 \mid L_p$ and $n \equiv p \pmod 4$ for every $p \mid n$ by Lemma 2.1. Thus all $p \equiv c$, and $n \equiv c^k \pmod 4$; if $c = 3$ this forces k odd. (b) For $p \neq 3$, $3 \mid p^2 - 1$, so $3 \mid L_p$ and $n \equiv p \pmod 3$. If $3 \mid n$ then every other prime factor would be $\equiv n \equiv 0 \pmod 3$, impossible for $k \ge 2$; so $3 \nmid n$, all $p \equiv c' \pmod 3$ and $n \equiv c'^k \equiv c'$, which for $c' = 2$ forces k odd. (c) is (a) and (b) with k even. □

Williams proved that $k = 3$ is impossible. For completeness we give a four-line proof in the present notation.

Proposition 2.3 (Williams)**.** *No CLC number has exactly three prime factors.*

Proof. Let $n = p_1p_2p_3$. By Lemma 2.1 at p_3, $p_1p_2 - 1 = tL_3$ with $t \geq 1$ (as $p_1p_2 > 1$); since $p_1p_2 < p_3^2$ we have $t < 2$, so $2p_1p_2 = p_3^2 + 1$. At p_2, $p_1p_3 - 1 = sL_2$ with $s \geq 1$; now $p_3 < \sqrt{2p_1p_2} < \sqrt{2}\,p_2$, so $p_1p_3 < \sqrt{2}\,p_2^2$ and $s < 2\sqrt{2}\,p_2^2/(p_2^2-1) < 3$. If $s = 2$ then $p_1p_3 = p_2^2$, impossible; so $2p_1p_3 = p_2^2 + 1$. Subtracting, $2p_1(p_2 - p_3) = p_3^2 - p_2^2$, i.e. $2p_1 = -(p_2 + p_3) < 0$, a contradiction. □

3. Four prime factors

3.1. **Structure.**

Theorem 3.1. *Let $n = p_1p_2p_3p_4$, $p_1 < p_2 < p_3 < p_4$, be a CLC number, and write $P = p_1p_2$, $q = p_3$, $r = p_4$.*

(a) *Every $p_i \equiv 1 \pmod{12}$; in particular $p_1 \geq 13$ and $p_2 \geq 37$.*

(b) *There are integers $d, e, s \geq 1$ with*

$$2Pq = d(r^2-1) + 2, \qquad 2Pr = e(q^2-1) + 2, \qquad 2p_1qr = s(p_2^2-1) + 2, \tag{1}$$

and consequently $q < 2P$, $r < \sqrt{2Pq} < 2P$, $m := qr < 4P^2$, $n < 4P^3$ and $d < 2p_1$.

(c) $p_2^2 \leq 24p_1^3 + 1$.

Consequently $p_2 < 4.9\,p_1^{3/2}$, $p_3, p_4 < 2P < 9.8\,p_1^{5/2}$ and $n < 477\,p_1^{15/2}$.

Proof. (a) is Lemma 2.2(c).

(b) Lemma 2.1 at r gives $n/r - 1 = Pq - 1 = dL_r$ with an integer $d \geq 1$, which is the first equation; the second and third are Lemma 2.1 at q and at p_2. From $2Pq = d(r^2-1)+2 \geq r^2+1 > q^2$ we get $q < 2P$ and $r^2 < 2Pq$, hence $r < \sqrt{2Pq} < 2P$, $m = qr < 4P^2$ and $n = Pm < 4P^3$. Also $d = 2(Pq-1)/(r^2-1) < 2Pq/(r^2-1) < 2P/q < 2p_1$ because $r \geq q+1$ and $q > p_2$.

(c) Multiply the three equations in (1). The left side is $8P^2p_1\,q^2r^2 = 8p_1^3p_2^2q^2r^2$. Writing $d(r^2-1)+2 = dr^2\bigl(1 - \frac{d-2}{dr^2}\bigr)$ and similarly for the other two factors, the right side is $d\,e\,s\;r^2q^2p_2^2\,\Pi$ with

$$\Pi = \Bigl(1 - \frac{d-2}{dr^2}\Bigr)\Bigl(1 - \frac{e-2}{eq^2}\Bigr)\Bigl(1 - \frac{s-2}{sp_2^2}\Bigr).$$

Hence

$$d\,e\,s \cdot \Pi = 8p_1^3. \tag{2}$$

For an integer $t \geq 1$ and $x \geq 2$ one has $-1/x^2 \leq (t-2)/(tx^2) < 1/x^2$, so each factor of Π lies in $[1 - 1/x^2,\, 1 + 1/x^2]$ with $x \in \{r, q, p_2\}$, $x \geq p_2$; therefore

$$(1 - p_2^{-2})^3 \leq \Pi \leq (1 + p_2^{-2})^3. \tag{3}$$

Now reduce the three equations (1) modulo p_1. Each left side is divisible by p_1, so $d(r^2-1) \equiv e(q^2-1) \equiv s(p_2^2-1) \equiv -2 \pmod{p_1}$. As p_1 is odd, $p_1 \nmid d$, $p_1 \nmid e$ and $p_1 \nmid s$. Hence $des \neq 8p_1^3$, i.e. $\Pi \neq 1$.

If $\Pi < 1$ then $des \geq 8p_1^3 + 1$ and $\Pi = 8p_1^3/des \leq 1 - \frac{1}{8p_1^3+1}$; with $\Pi \geq (1 - p_2^{-2})^3 \geq 1 - 3p_2^{-2}$ this gives $p_2^2 \leq 3(8p_1^3+1) = 24p_1^3 + 3$. If $\Pi > 1$ then $des \leq 8p_1^3 - 1$ and $\Pi \geq 1 + \frac{1}{8p_1^3-1}$; since $p_2 \geq 37$, $(1+p_2^{-2})^3 \leq 1 + 3.0001\,p_2^{-2}$, which gives $p_2^2 \leq 3.0001(8p_1^3-1) < 24p_1^3 + 3$. In both cases $p_2^2 \leq 24p_1^3 + 3$. Finally $p_2 \equiv 1 \pmod{12}$ implies $p_2^2 \equiv 1 \pmod{24}$, while $24p_1^3 + 2$ and $24p_1^3 + 3$ are $\equiv 2, 3 \pmod{24}$; so $p_2^2 \leq 24p_1^3 + 1$.

The consequences follow from (b) and (c): $p_2 < \sqrt{24}\,p_1^{3/2} < 4.9\,p_1^{3/2}$, $q, r < 2P < 9.8\,p_1^{5/2}$, and $n < 4P^3 < 4 \cdot 4.9^3\,p_1^{15/2} < 477\,p_1^{15/2}$. □

Remark 3.2. In fact $\Pi < 1$ always: a factor of Π exceeds 1 only when the corresponding quotient equals 1, and $e, s \geq 2$ (as $e > d \geq 1$ by $r > q$, and $s = 2(p_1qr-1)/(p_2^2-1) > 2p_1$), so $\Pi > 1$ would need $1/r^2 > (1-2/e)/q^2 + (1-2/s)/p_2^2$ up to second-order terms, which fails for $r > q > p_2$. We kept both cases in the proof to avoid relying on this.

Corollary 3.3. *For each prime p_1 there are only finitely many four-prime CLC numbers with smallest prime p_1, all below $477\,p_1^{15/2}$. Conversely a four-prime CLC number n has $p_1 > (n/477)^{2/15}$.*

3.2. **Enumeration.** Theorem 3.1 makes two kinds of complete enumeration possible: all $n \leq B$ (where $n > p_1p_2^3$ forces $p_1 < B^{1/4}$ and $p_2 < (B/p_1)^{1/3}$), and all n whatsoever with $p_1 \leq X$ (where $p_2 \leq \sqrt{24p_1^3+1}$). Both reduce to a search over pairs (p_1, p_2).

Lemma 3.4 (one class for the cofactor)**.** *With notation as in Theorem 3.1, $m = qr$ satisfies $p_2m \equiv 1 \pmod{L_{p_1}}$ and $p_1m \equiv 1 \pmod{L_{p_2}}$. These are solvable iff $p_1 \equiv p_2 \pmod{\gcd(L_{p_1}, L_{p_2})}$, in which case m lies in a single residue class modulo $\Lambda = \operatorname{lcm}(L_{p_1}, L_{p_2})$.*

Proof. The congruences are Lemma 2.1 at p_1 and p_2 ($n/p_1 = p_2m$, $n/p_2 = p_1m$). Let $g = \gcd(L_{p_1}, L_{p_2})$. The system is solvable iff $p_2^{-1} \equiv p_1^{-1} \pmod g$, i.e. $p_1 \equiv p_2 \pmod g$ (both p_i are coprime to g: $p_2 \mid L_{p_1}$ would need $p_2 \mid p_1 \pm 1$ with $p_2 > p_1$, impossible for odd primes, and $p_1 \mid L_{p_2}$ would make $p_1m \equiv 1 \pmod{p_1}$ impossible). The rest is the Chinese remainder theorem. □

Since $\Lambda \approx P^2/(4g)$ while $m < 4P^2$, Lemma 3.4 leaves $O(g)$ candidates m per pair. The next lemma rejects almost all of them with one multiplication.

Lemma 3.5 (window)**.** *With d, e as in (1) and $j := de$, $T := 4P^2$,*

$$|jm - T| < \max\bigl(2j\sqrt{2p_1},\ 8p_1\bigr). \tag{4}$$

Moreover, for fixed $m > 2\lceil\sqrt{2p_1}\rceil$ the set of real $j > 0$ satisfying $|jm-T| \leq \max(2j\lceil\sqrt{2p_1}\rceil, 8p_1)$ is an interval containing T/m; hence it contains an integer iff it contains $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$.

Proof. From (1), $d = 2(Pq-1)/(r^2-1)$ and $e = 2(Pr-1)/(q^2-1)$, so $j\,(q^2-1)(r^2-1) = 4(Pq-1)(Pr-1)$. Expanding with $m = qr$,

$$jm^2 - Tm = j(q^2+r^2-1) - 4P(q+r) + 4.$$

The right side is a difference of two positive numbers, so its absolute value is at most $\max\bigl(j(q^2+r^2-1),\ 4P(q+r)-4\bigr) \leq \max(2jr^2,\ 8Pr)$. Dividing by $m = qr$ gives $|jm-T| \leq \max(2jr/q,\ 8P/q)$. By Theorem 3.1(b), $r^2 < 2Pq$, so $r/q < \sqrt{2P/q} < \sqrt{2p_1}$ (as $q > p_2$), and $P/q < p_1$; this is (4). For the interval property, put $S = \lceil\sqrt{2p_1}\rceil$ and $\phi(j) = |jm - T| - \max(2jS, 8p_1)$. On $j \geq T/m$ the first term increases with slope m and the second with slope at most $2S < m$, so ϕ is strictly increasing; on $j \leq T/m$ the first term decreases and the second is non-decreasing, so ϕ is strictly decreasing. Thus $\{\phi \leq 0\}$ is an interval, and it contains T/m since $\phi(T/m) < 0$. □

Algorithm. For every pair of primes $p_1 < p_2$, both $\equiv 1 \pmod{12}$, within the range dictated by the campaign, (i) test the compatibility condition of Lemma 3.4 and compute m_0 and Λ; (ii) walk the candidates $m \equiv m_0 \pmod\Lambda$ in $(p_2^2, \min(4P^2, B/P)]$, rejecting m unless $\lfloor T/m\rfloor$ or $\lceil T/m\rceil$ satisfies (4) (for large m the walk is organised over the integers j instead, which is equivalent by Lemma 3.5 and much faster); (iii) for each survivor, require $2^{n-1} \equiv 1 \pmod n$ (necessary for a Carmichael number); (iv) for each remaining m, recover r as an integer root of $dx^3-(d-2)x-2n$ for $d = 1, \ldots, 2p_1$ (multiply the first equation of (1) by r: $2n = dr^3-(d-2)r$), set $q = m/r$, and check all four conditions of Lemma 2.1 together with the primality of q and r. Every survivor of step (ii), and every reported solution, is additionally re-verified by an independent Python program using exact integer arithmetic and SymPy factorisation.

Validation. The implementation was tested against brute force on a *relaxed* problem that does have solutions: replace L_p by $(p^2-1)/\kappa$ for $\kappa \in \{24, 120\}$. The whole derivation above goes through with 2 replaced by κ. On the relaxed problem the enumerator finds exactly the solution set that a direct four-fold loop finds (for $\kappa = 24$ the unique solution below 10^{12} is $13685 = 5 \cdot 7 \cdot 17 \cdot 23$, for which $d = 27$, $e = 67$, $s = 977$ and $des \cdot \Pi = 24^3 \cdot 5^3$ exactly). Separate self-tests brute-force the two-prime sub-problem (q, r) for small (p_1, p_2) in four relaxed variants (43 solutions) and check every inequality used above, and brute-force the residue class of Lemma 3.4 for 13,037 pairs. On the real problem the enumerator reproduces the empty result below 10^{24} of Theorem 4.1 in six seconds. One design error was caught by these tests before any production run: the true $j = de$ is *not* in general $\lfloor T/m \rfloor$ or $\lceil T/m \rceil$ (only the filter may use those two values), so the reconstruction step (iv) must not depend on j.

Theorem 3.6. *There is no CLC number with exactly four prime factors and $n \le 10^{30}$, and there is none at all whose smallest prime factor is at most* 500,000*. Consequently a four-prime CLC number, if one exists, has $p_1 >$ 500,000 and $n > 10^{30}$.*

Computation. Table 1 summarises the two campaigns (Apple M4 Pro, 14 threads). In each, the single survivor of the Fermat filter is the Carmichael number $26921089 = 13 \cdot 37 \cdot 97 \cdot 577$, for which $38 \nmid 26921090$; all window survivors with $P > 2^{40}$ were re-verified in Python. The second statement combines the second campaign with Corollary 3.3. □

campaign	pairs (p_1, p_2)	window survivors	Fermat survivors	solutions	wall time
all $n \le 10^{30}$	230,612,582,193	1,904	1	0	87 min
all n, $p_1 \le$ 500,000	86,517,004,140	1,185	1	0	146 min

TABLE 1. The four-prime enumeration (Theorem 3.6).

4. No CLC number below 10^{24}

Theorem 4.1. *No CLC number is smaller than 10^{24}. Among the* 308,279,939 *Carmichael numbers below 10^{24}, exactly one has four prime factors p with $(p+1) \mid (n+1)$, namely*

$$15500309690554865758009 = 37 \cdot 103 \cdot 109 \cdot 127 \cdot 307 \cdot 523 \cdot 613 \cdot 1429 \cdot 2089$$

(the primes $37, 109, 1429, 2089$), 826 have exactly three, and none has more than four. Below 10^{22} the maximum is three, attained by 179 *numbers.*

Method. Carmichael numbers have been tabulated completely to 10^{21} by Pinch [14, 15], to 10^{22} by Goutier [9] and to 10^{24} by Shallue and Webster [17, 18], whose list is distributed by Stamm-Wilbrandt as a PARI/GP binary vector [19]. A CLC number is in particular Carmichael, so it suffices to test every tabulated Carmichael number for the Lucas–Carmichael property. We did this ourselves rather than rely on flags computed by the tabulators: each n was factored (the Miller–Rabin square-root-of-unity splitting, which always succeeds on Carmichael numbers, with a general factoring fallback), the factorisation was checked to be squarefree with product n and to satisfy Korselt's criterion, and the number of primes p with $(p+1) \mid (n+1)$ was recorded. Processing is line-based so that no entry can be skipped at a buffer boundary. Three completeness checks were applied to the 10^{24} list: the counts below 10^{22}, 10^{23} and 10^{24} (49,679,870; 123,381,982; 308,279,939) equal the published ones [18], the list is strictly increasing, and it begins $561, 1105, 1729, 2465, 2821$. Our per-$\omega$ tallies agree with independently published counts: 1,494,738 three-prime Carmichael numbers below 10^{24} (OEIS A132195, [6]) and 6,529,921 eleven-prime ones (OEIS A299711); below 10^{22} all eleven per-ω counts agree with Goutier's tally, and

the 179 three-match numbers agree with Eldar's A329948. The converse direction was also checked: none of the 129,125 Lucas–Carmichael numbers below 10^{15} tabulated by Suteu (OEIS A006972) is Carmichael. Table 2 gives the distribution below 10^{24}.

$\omega(n)$	count	$\omega(n)$	count	$\omega(n)$	count
3	1,494,738	7	62,338,997	11	6,529,921
4	548,265	8	99,469,866	12	547,590
5	3,429,390	9	81,041,495	13	15,328
6	19,665,194	10	33,199,094	14	61

TABLE 2. Carmichael numbers below 10^{24} by number of prime factors, as recounted by our scan (total 308,279,939).

Corollary 4.2. (a) *A counterexample to Agrawal's conjecture of the Lenstra–Pomerance type (a CLC number all of whose primes are* $\equiv 3 \pmod{80}$*) exceeds* 10^{24}.

(b) *A Baillie–PSW counterexample of the Pomerance type (a Carmichael number with* $(p+1) \mid (n+1)$ *for all* $p \mid n$*) exceeds* 10^{24}.

Statement (b) adds little to the known absence of Baillie–PSW counterexamples below $2^{64} \approx 1.8 \times 10^{19}$ [8], but it is a different kind of statement: it bounds every construction that proceeds through the Pomerance conditions, not only the tested range.

Remark 4.3 (Howe's order-2 Carmichael numbers). Howe [11] calls n a Carmichael number of order 2 if $n \equiv p \pmod{p^2-1}$ for all $p \mid n$; by Lemma 2.1 a CLC number is the same with modulus $(p^2-1)/2$. Thus a CLC number is of order 2 iff additionally $(n-p)/L_p$ is even for every $p \mid n$. Tabulations of order-2 Carmichael numbers therefore bound one 2-adic type of CLC number; we have not pursued this.

5. Obstructions to constructions

Carmichael numbers are constructed in two ways: by polynomial families in the manner of Chernick [5], $n(k) = (6k+1)(12k+1)(18k+1)$, where the divisibilities are polynomial identities; and by subset products in the manner of Erdős and Alford–Granville–Pomerance [7, 2, 12], where many primes p with $p-1$ dividing a fixed smooth Λ are harvested and a subset with product $\equiv 1 \pmod{\Lambda}$ is sought. We show that the two-sided versions of both break down in their simplest forms.

5.1. **Polynomial families.** Consider $n(k) = \prod_{i=1}^{m}(a_i k + b_i)$ with integers $a_i > 0$, the linear factors pairwise non-proportional, and require the CLC divisibilities as polynomial identities: $(a_i k + b_i - 1) \mid (n(k) - 1)$ and $(a_i k + b_i + 1) \mid (n(k) + 1)$ in $\mathbb{Q}[k]$. Since a linear polynomial divides f iff f vanishes at its root, this is $n\big((1-b_i)/a_i\big) = 1$ and $n\big((-1-b_i)/a_i\big) = -1$. With $D_{ij} = a_i b_j - a_j b_i$ one computes $a_j \frac{1-b_i}{a_i} + b_j = (a_j + D_{ij})/a_i$, so the conditions read, for every i,

$$\prod_{j \neq i}(a_j + D_{ij}) = a_i^{m-1} = \prod_{j \neq i}(D_{ij} - a_j). \tag{5}$$

Theorem 5.1 (three factors)**.** *No such family exists for* $m = 3$.

Proof. Subtracting the two products in (5) for $i = 1, 2, 3$ gives $a_2 D_{13} + a_3 D_{12} = 0$, $a_1 D_{23} + a_3 D_{21} = 0$, $a_1 D_{32} + a_2 D_{31} = 0$, i.e. $a_2 D_{13} = -a_3 D_{12}$, $a_1 D_{23} = a_3 D_{12}$, $a_1 D_{23} = -a_2 D_{13}$. The identity $a_1 D_{23} - a_2 D_{13} + a_3 D_{12} = 0$ (it is $a \wedge b \wedge a = 0$) becomes $3 a_3 D_{12} = 0$, so $D_{12} = 0$ and then $D_{13} = D_{23} = 0$: b is proportional to a and the factors are proportional, contrary to hypothesis. □

Theorem 5.2 (uniform offsets)**.** *No such family exists with $b_i = -1$ for all i, nor with $b_i = +1$ for all i, for any $m \geq 2$.*

Proof. If all $b_i = -1$ then $D_{ij} = a_j - a_i$, the second product in (5) is $(-a_i)^{m-1}$, and the first becomes $\prod_{j\neq i}(2a_j - a_i) = a_i^{m-1}$. The a_i are distinct (non-proportionality). Take i with a_i maximal: for $j \neq i$, $0 < a_j < a_i$ gives $|2a_j - a_i| < a_i$, so the product has absolute value $< a_i^{m-1}$. The case $b_i = +1$ is the mirror image with $\prod_{j\neq i}(a_i - 2a_j)$. □

Theorem 5.3 (generic position)**.** *Put $s_i = (b_i - 1)/a_i$ and $t_i = (b_i + 1)/a_i$. If the s_i are pairwise distinct and the t_i are pairwise distinct, no such family exists, for any $m \geq 2$.*

Proof. Let $G(\tau) = n(-\tau)$, of degree m with leading coefficient $c = (-1)^m \prod a_i$. The hypotheses say $G(s_i) = 1$ and $G(t_i) = -1$. With m distinct roots each, $G - 1 = c\prod(\tau - s_i)$ and $G + 1 = c\prod(\tau - t_i)$. Subtracting, $2 = c\big[\prod(\tau - t_i) - \prod(\tau - s_i)\big]$ identically in τ. The coefficient of τ^{m-1} on the right is $-c\sum(t_i - s_i) = -c\sum 2/a_i \neq 0$, while on the left it is 0. □

Corollary 5.4. *Every CLC polynomial family has two indices $i \neq j$ with $p_j - 1 = \frac{a_j}{a_i}(p_i - 1)$ or $p_j + 1 = \frac{a_j}{a_i}(p_i + 1)$ as polynomial identities.*

The degenerate strata left open by Theorem 5.3 were searched exhaustively using the fact that each factor in (5) divides a_i^{m-1}: no family exists with $m = 4$ and all $a_i \leq 500$, with $m = 5$ and $a_i \leq 150$, or with $m = 6$ and $a_i \leq 60$ (45 min, 1.8 h and 54 h respectively). Theorem 5.2 was also confirmed by search for $m = 5, 7, 9, 11$ up to $a_i \leq 20000, 5000, 3000, 1500$.

Remark 5.5. These theorems close families in which the divisibilities hold identically in k. They say nothing about families that are CLC only for k in some residue classes, nor about sporadic solutions. Theorem 5.1 is the parametric shadow of Proposition 2.3.

5.2. **Two-sided subset products.** In the subset-product method one fixes a smooth modulus Λ, harvests primes p with $(p - 1) \mid \Lambda$, and looks for a subset with product $\equiv 1 \pmod{\Lambda}$; the expected number of solutions is about $2^{(\text{pool size}) - \log_2 \varphi(\Lambda)}$, and the method succeeds because the pool grows faster than $\log_2 \varphi(\Lambda)$. For CLC numbers one needs simultaneously $(p + 1) \mid \Lambda'$ and product $\equiv -1 \pmod{\Lambda'}$, and the two moduli interact: by Lemma 2.2 an odd prime ℓ cannot divide both some $p_i - 1$ and some $p_j + 1$ (else $n \equiv 1 \equiv -1 \pmod{\ell}$), so every small prime must be assigned to one side, and the 2-adic branch is fixed. Define the *margin* of a configuration as (number of usable primes) $- \log_2 \varphi(\text{lcm of all required } p \mp 1)$, so that the expected number of two-sided solutions is about 2^{margin}.

We measured the margin over three families of configurations: random smooth moduli (3,250 configurations, all with margin > -14 searched exhaustively by meet-in-the-middle, no solution); pools of all primes $p \leq X$ with both $p-1$ and $p+1$ y-smooth, with the side assignment optimised by hill-climbing (hundreds of restarts); and pools restricted to primes with few prime factors in $p \mp 1$. The best margins were -8.2 (tiny three-prime systems), $-29.0/-30.6/-31.0/-36.9$ for $y = 60/100/140/200$, and -36.9 for the sparse pools. The wall is flat or worsening in every direction we varied: each usable prime contributes one bit of subset freedom but about 2.5 bits of modulus, because a prime that is useful on one side is almost never useful on the other. This is a measurement, not a theorem, but it quantifies the "intractable" judgement of the Baillie–PSW literature [3] and explains it.

6. Heuristics

Four prime factors. Condition on Theorem 3.1(a). For $p \equiv 1 \pmod{12}$ the condition $n/p \equiv 1 \pmod{L_p}$ is automatic modulo 12, so model the remaining condition as an event of probability $24/p^2$, independently for the four primes. Summing $\prod_i 24/p_i^2$ over quadruples of primes $\equiv 1$

(mod 12) with $p_2^2 \le 24p_1^3 + 1$ and $p_2 < q < r < 2p_1p_2$ (Theorem 3.1) gives an expected count of about 7.5×10^{-7}, of which 97% comes from $p_1 = 13$ (where $n < 1.1 \times 10^{11}$, excluded since [14]); the contribution of $p_1 > X$ decays like X^{-4} and is below 10^{-25} for $X = 500{,}000$. On this model the four-prime case is empty with overwhelming probability, and Theorem 3.6 confirms the part of the prediction that can be tested.

Onset of CLC numbers. An Erdős–Pomerance-style count of k-factor products of primes with $p \mp 1$ both y-smooth, including the side-consistency factor $2(1-1/\ell)^k - (1-2/\ell)^k$ for each odd prime $\ell \le y$ and a factor 2^{1-k} for the 2-adic branch, calibrated on the one-sided side against Pinch's counts $C(10^{15}) = 105{,}212$, $C(10^{18}) = 1{,}401{,}644$ and on the two-sided side against our measured counts of twin-smooth primes, places the first CLC number near 10^{208}–10^{490} with $k \approx 80$–110 and $y \approx 40$–90, and the first Lenstra–Pomerance counterexample to Agrawal's conjecture near 10^{2400}–10^{2900} with $k \approx 270$. The ranges reflect the sensitivity to the per-prime calibration; the qualitative conclusion—far beyond exhaustive search, reachable only by a construction that circumvents §5—is robust. The same machinery places the first Carmichael number near 10^3 (truth: 561).

7. Reproducibility

All code is in the public repository [**TODO: URL**], tagged for this version. The 10^{24} scan (`clc/`) consists of the download–decode–verify–scan pipeline, the completeness report, the result files with per-ω and per-match counts, and SHA-256 checksums of the input data. The four-prime enumeration (`clc/k4/`) consists of the C program, its brute-force reference, the Python re-verifier, the per-block logs of both campaigns (one JSON line per block, so the coverage can be audited), and `make test`, which runs the self-tests and the relaxed end-to-end comparison in under a minute. The family searches and the heuristic model are in `agrawal/`, with a PYTEST suite that includes symbolic checks of the identities used in the proofs of Theorems 3.1 and 5.1–5.3. The consultation history, including the errors caught along the way, is in `docs/`.

8. Open problems

(1) Is $k = 4$ impossible? Theorem 3.1 reduces it to one free parameter; the integer $\delta = des - 8p_1^3 \ge 1$ equals $8p_1^3(1/\Pi - 1)$, which pins the sizes of q, r, p_2 relative to p_1 but does not by itself exclude them.
(2) The $k = 5$ case, where both classes modulo 4 are allowed and the analogue of (2) has four factors.
(3) Closing the $m = 4$ polynomial-family stratum with one repeated root.
(4) A rigorous two-sided analogue of Pomerance's upper bound for Carmichael numbers.

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